MILITARY SPECIFICATION

CONTROL UNIT, COMMUNICATION SYSTEM C-11746(V)( )/ARC

This specification is approved for use by the US Army Aviation Systems Command, U.S. Army Avionics Research and Development Activity, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of control-unit which provides control of intercommunication, audio levels, switching of airborne radio sets and navigation devices and is designated Control Unit, Communication System C-11746(V)( )/ARC. (See 6.1 and 6.4.)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

NN-P-71	Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
QQ-S-781	Strapping, Steel, and Seals.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Avionics Research & Development Activity, ATTN: SAVAA-C, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5895

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock Corner.
PPP-B-636	Box, Shipping Fiberboard.
PPP-B-640	Boxes, Fiberboard, Corrugated, Triplewall.
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-S-760	Strapping, Nonmetallic (and Connectors).
PPP-T-97	Tape, Pressure-Sensitive, Adhesive, Filament Reinforced.

MILITARY

MIL-P-116	Preservation Methods of.
MIL-E-5400	Electronic Equipment, Aerospace, General Specification for.
MIL-F-7179	Finishes, Coatings and Sealants for the Protection of Aerospace Weapons Systems.
MIL-P-11268	Parts, Materials and Processes Used in Electronic Equipment.
MIL-M-13231	Marking of Electronic Items.
MIL-S-19500	Semiconductor Device, General Specification for.
MIL-P-19644	Plastic Molding Material (Polystyrene Foam, Expanded Bead).
MIL-I-23053	Insulation Sleeving, Electrical, Heat Shrinkable, General Specification for.
MIL-C-25050	Colors, Aeronautical Lights and Lighting Equipment, General Requirements for.
MIL-L-25467	Lighting, Integral Red Aircraft Instrumentation, General Specification for.
MIL-P-26514	Polyurethane Foam, Rigid or Flexible for Packaging.
MIL-P-55110	Printed Wiring Boards, General Specification for.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

MIL-I-58110(AV)	Indicator, Remote ID-2403/ARC

STANDARDS

FEDERAL

FED-STD-595	Color Requirements for Individual Color Chips.

MILITARY

MIL-STD-105	Sampling Procedures and Tables of Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-147	Palletized Unit Loads.
MIL-STD-252	Classification of Visual and Mechanical Defects.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-461 (Notice 4)	Electromagnetic Interference Characteristics Requirements for Equipments.
MIL-STD-462 (Notice 2)	Electromagnetic Interference Characteristics, Measurements of.
MIL-STD-704	Aircraft, Electric Power Characteristics.
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production.
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MS25212	Control Panel, Console Type, Aircraft Equipment, Basic Dimensions.

2.1.2 Other Government documents, drawings and publications. The following government documents form a part of this specification to the extent specified herein.

DRAWINGS

Avionics Research and Development Activity (FSCM 57045)

D6002801	Control Unit, Communication System C-11746(V)1/ARC
D6002802	Control Unit, Communication System C-11746(V)2/ARC

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

D6002803	Control Unit, Communication System C-11746(V)3/ARC
D6002804	Control Unit, Communication System C-11746(V)4/ARC
D6002805	Panel Illuminated RED, 28V MX-10844/ARC
D6002806	Panel Illuminated RED, 5V MX-10845/ARC
D6002807	Panel Illuminated GRN, 115V MX-10846/ARC
D6002808	Panel Illuminated GRN, 28V MX-10847/ARC
D6002810	Control Unit, Communication System C-11747(P)/ARC

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

PUBLICATIONS

The following document is classified but is available for review at CECOM, Fort Monmouth, NJ upon request through the Contracting Officer for those having proper security clearances on file at CECOM Security Office:

NACSEM-5100	Compromising Emanations Laboratory Test Standard, Electromagnetics (Document CONFIDENTIAL Title Unclassified).

2.1.3 Specifications of exact issue. The following documents of the exact issue shown form a part of this specification to the extent specified herein. In the event of conflict between the documents referenced herein and the contents of this specification, the contents of this specification shall be considered a superseding requirement.

MIL-STD-781B	Reliability Design Qualification and Production Acceptance Tests; Exponential Distribution.
MIL-STD-810C	Environmental Test Methods and Engineering Guidelines.
MIL-L-85762	Lighting, Aircraft, Interior, AN/AVS-6 Aviator's Night Vision Imaging System (ANVIS) Compatible.
MIL-P-7788E	Panels, Information, Integrally Illuminated.

2.1.4 Other Publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-government documents which is current on the date of the solicitation.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:19Z
Check the source to verify that this is the current version before use.

AMERICAN SOCIETY OF TESTING MATERIALS

D-3951 Jan 82 Commercial Packaging

(Application for copies should be addressed to the American Society for Testing Materials (ASTM), 1916 Race Street, Philadelphia, PA 19103.

(Non-government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations uless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The Control Unit, Communication System C-11746(V)( )/ARC is a lightweight, self-contained, solid state unit containing switching/mixing/amplifying circuitry designed to process audio output to an individual headset from up to five communications receivers and six navigation receivers. Each of the seven monitor switches has an integral audio level adjustment. A main volume control provides simultaneous audio level adjustment of the eleven selectable receiver circuits plus two direct inputs. Two additional direct audio inputs are provided without audio level control. An eight position switch selects one of up to five radio sets for transmission and reception, a private (PVT) intercom line, or an interphone (ICS) position for use when two or more controls are interconnected within an airframe, or the Remote Select Mode (RMT). A four-position rotary switch permits operating the ICS line in HOT-MIC, or VOX (voice activated mic), or NORM, or disabling the ICS audio input. A two-position flush-mounted, thumb-actuated switch is used to select either a 5 ohm or a 150 ohm microphone. A VOX control sets the threshold for VOX operation. The unit is flush-mounted by means of Dzus fasteners located on the sides of the front panel mounting plate; operating controls and panel lamps are mounted on the front panel. The unit is capable of operating from a power source of 27.5 volts dc nominal. Panel lighting is operable from either +5 volts dc, 5 volts ac 400 Hz, 115 volt ac 400 Hz, or 27.5 volt dc nominal depending upon panel selection. All field level adjustments and serviceable items shall be accessible without removal of cover plates and/or panels.

3.1.1 Remote Control. The unit shall be capable of accommodating two types of remote control. In the sequential mode, transmit selection shall be by means of an external momentary contact switch. In the SP5T mode, transmit selection shall be accomplished by actuating one of five momentary contact switches, one for each radio channel. Operation of one of the switches shall also key the selected radio. An SP5T switch has five positions, namely; up, down, left, right, center-in.

3.1.2 Piggy Back Configuration. The unit shall be capable of being configured in tandem with another unit to provide selection and control of 10 radios at a single station. This configuration shall be accomplished by external connections only. Remote sequential mode control shall be operable in this configuration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

3.1.3 VOX. The unit shall be capable of voice actuated mic keying for the ICS channel only. This feature will be selectable from the front panel HOT MIC/VOX ON/ NORM/ICS OFF switch. In addition, a VOX threshold control shall be available on the front panel.

3.2 Construction. The units on order shall be constructed in accordance with the following drawing and data lists:

DRAWINGS

D6002801	Control Unit, Communication System C-11746(V)1/ARC
D6002802	Control Unit, Communication System C-11746(V)2/ARC
D6002803	Control Unit, Communication System C-11746(V)3/ARC
D6002804	Control Unit, Communication System C-11746(V)4/ARC
D6002805	Panel Illuminated RED, 28V MX-10844/ARC
D6002806	Panel Illuminated RED, 5V MX-10845/ARC
D6002807	Panel Illuminated GRN, 115V MX-10846/ARC
D6002808	Panel Illuminated GRN, 28V MX-10847/ARC
D6002810	Control Unit, Communication System, C-11747(P)/ARC

3.2.1 Weight (see 4.14.1). The weight of the completely assembled unit shall not exceed 2.6 pounds.

3.2.2 Outline dimensions (see 4.14). The dimensions of the completely assembled unit shall conform to the requirements of Military Standard MS 25212 (ASG) within the following limitations:

Panel/Housing Height - 2.625 inches (6.67 cm)

NOTE: Housing height behind front panel such that dimension .048 min. between top of panel and top of housing and between bottom of panel and bottom of housing.

Panel Width - 5.75 inches (14.60 cm)

Housing Width - 5.0 inches max. (12.77 cm)

Housing Depth - 5.12 inches (11.75 cm) (excluding front panel and rear connectors)

3.3 First article. When specified in the contract or purchase order, the contractor shall furnish First Article units as required (see 4.3 and 6.2d).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

3.4 Finish (see 4.4). The case of the control shall be black, Color Number 37038 of FED-STD-595. The finishes and protective coatings used on all parts shall be in accordance with MIL-E-5400 and MIL-F-7179. Protective coatings shall be field restorable. All parts shall be suitably treated or finished so as to provide protection from corrosion in accordance with 3.1.2 of MIL-F-7179.

3.5 Securing of parts (see 4.14).

3.5.1 General. Brackets, lugs, inserts, bolts, and other mounting arrangements shall retain items securely when subjected to specified service conditions.

3.6 Marking (see 4.4).

3.6.1 General. Marking shall conform to MIL-M-13231. Front panel marking shall be in accordance with equipment drawings.

3.6.2 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.6.3 Serial numbers. The following unit(s) shall be serial numbered.

DRAWINGS

D6002801	Control Unit, Communication System C-11746(V)1/ARC
D6002802	Control Unit, Communication System C-11746(V)2/ARC
D6002803	Control Unit, Communication System C-11746(V)3/ARC
D6002804	Control Unit, Communication System C-11746(V)4/ARC
D6002805	Panel Illuminated RED, 28V MX-10844/ARC
D6002806	Panel Illuminated RED, 5V MX-10845/ARC
D6002807	Panel Illuminated GRN, 115V MX-10846/ARC
D6002808	Panel Illuminated GRN, 28V MX-10847/ARC
D6002810	Control Unit, Communications System, C-11747(P)/ARC

3.6.4 Front panels (green). (See 4.9.2)

3.6.4.1 Panel contruction. The unit(s) shall have an integrally illuminated front information panel. The panel shall have white legends against a black background for daylight operation and shall have green primary lighting for night operation, to provide compatibility with night vision goggles. Panel fabrication and perfonmance shall be in accordance with Class 1-R, (except that the color is green) Type VI of MIL-P-7788E. The panel lettering font shall be Alternate Gothic No. 3.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:19Z
Check the source to verify that this is the currentversion before use.

3.6.4.1.1 Chromaticity and spectrum. Normal legend illumination shall be ANVIS GREEN A with a U' coordinate of .099, V' coordinate of .544 and a radius of .034 on the 1976 UCS chromaticity diagram measured at 0.1 Ft. Lambert display luminance. The spectral radiance shall be such that the ANVIS RADIANCE (AR) shall not exceed 1.7 x 10^{-10} AR when illuminated to produce 0.1 Ft. Lambert (FL) display luminance when measured as per the requirements of MIL-L-85762.

3.6.4.1.2 Brightness. The brightness shall be 1.0 ± 0.5 Foot-Lamberts at nominal panel supply voltage.

3.6.4.2 Power. Panel lighting shall be operable from the 28 volt dc or 5 volt dc/ac (400 Hz) aircraft source. Green panel lighting equipment shall also be operable from the 115 volt ac (400 Hz) or 28 volt dc aircraft source.

3.6.4.3 Panel Protection. The CSC shall be designed so that each panel type, i.e., 5 volt, 28 volt, or 115 volt, receives its power over a separate power input pin, both at the input connector J1 and, at the panel connector. This is required to prevent damage to the CSC should it inadvertently be installed in the incorrect aircraft.

The following pins are designated for panel illumination in input connector J1:

Panel Illumination Voltage	Pin Number
5 volts	R
28 volts	E
115 volts	J
Common	U

3.6.5 Alternate front panel red. (See 4.9.2). This paragraph shall be implemented only if contractually required. The units shall have an integrally illuminated front information panel. The panel shall have white legends against a black background for daylight operation and shall have red lighting for night operation. Panel fabrication and performance shall be in accordance with Class 1-R, Type V of MIL-P-7788E. All lighting shall be aviation-instrument panel lighting red (@.620) as specified in MIL-C-25050 and shall meet the requirements of MIL-L-25467, MIL-P-7788 and MIL-STD-1472 and shall be externally controllable. By a change of panel, the unit(s) shall be compatible with either a nominal 28 volt dc or a nominal 5 volt dc/ac (400 Hz) console panel lighting bus.

3.6.6 ICS switch. The HOT MIC/VOX ON/NORM/ICS OFF Switch shall be a rotary 4 position switch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

3.6.7 Monitor switches. The integral monitor switch/level control shall be a combination push-pull (on-off) switch and rotary potentiometer. The switch shall be configured such that depressing the switch will activate that monitor line and pulling out on the switch will inactivate the monitor line. Minimum volume will be achieved at full counter-clockwise rotation of the potentiometer and maximum volume at full clockwise rotation.

3.6.8 Interface wiring. The assignment of interface connector pins shall be as specified in table 1. All radio and interphone audio inputs and outputs of the unit are designed for balanced operation with respect to signal ground, using twisted, shielded pair wiring. The unit shall be designed so that a single transmit-receive set unbalance occurring in the system wiring (audio lows shorted to ground) will not deleteriously impact on the operational and isolation integrity of the unit.

3.7 Parts, materials, and processes; general (see 4.14). In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply.

3.8 Printed wiring assemblies (see 4.4). Printed wiring assemblies shall be processed for compliance with the requirements specified on the equipment drawings and MIL-P-55110 in order not to affect life and serviceability of the assemblies.

3.9 Semiconductors (transistors and diodes) (see 4.4). These electronic devices (semiconductors and diodes) shall conform to MIL-S-19500 and the equipment drawings.

3.10 Tropicalization of materials (see 6.2.g). Materials shall be treated as follows:

3.10.1 Treating materials. Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.10.2 Toxicity. Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.10.3 Flexibility. Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.11 Environmental service conditions. The equipment shall meet the operating and nonoperating service condition requirements of MIL-E-5400 for Class 1A equipment, except as specified below. The equipment shall be operable without degradation in specified performance before, during and after environmental testing unless otherwise specified herein. In the event of a relevant failure during an environmental test necessitating a corrective action, the test will be repeated upon completion of the repair or modification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

TABLE I. Pin connections

J1 D6001897-5 Mates with MS 3126E20-41S	
Connector Pin	Function
C	28 VDC Power
D	Power Ground
R	Panel Light +5V
U	Panel Light Power Ground
e	Chassis Ground
H	Chassis Ground
L	Microphone Input (HI)
M	Microphone Input (LO)
K	Wideband Audio Out
b	Interphone Key
c	Key Common
d	Radio Key
f	Transmitter Control #1
g	Transmitter Control #2
h	Transmitter Control #3
r	Transmitter Control #4
t	Transmitter Control #5
X	Logic Select (PVT)
Y	Logic Select (ICS)
Z	Logic Select (XMIT 1)
i	Logic Select (XMIT 2)
j	Logic Select (XMIT 3)
m	Logic Select (XMIT 4)
n	Logic Select (XMIT 5)
N	Transmit Audio Out 1A
P	Transmit Audio Out 1B
S	Transmit Audio Out 2A
T	Transmit Audio Out 2B
V	Transmit Audio Out 3A
W	Transmit Audio Out 3B
k	Transmit Audio Out 4A
s	Transmit Audio Out 4B
p	Transmit Audio Out 5A
q	Transmit Audio Out 5B
A	PVT Hot Mic
B	Remote Xmit Sel 5
E	Panel Light +28V
F	Remote Xmit Sel 4
G	Aux Mic In
J	Panel Light 115V AC
a	Remote Xmit Sel 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

TABLE I. Pin connections - Continued

J2 D6001897-6 Mates with MS 3126E20-41SW	
Connector Pin	Function
C	Chassis Gnd
D	Headset Audio (HI)
E	Headset Audio (LO)
g	Private Line PVT-A
P	Private Line PVT-B
R	Interphone Line ICS-A
S	Interphone Line ICS-B
A	Receiver Audio In 1A
B	Receiver Audio In 1B
V	Receiver Audio In 2A
W	Receiver Audio In 2B
U	Receiver Audio In 3A
j	Receiver Audio In 3B
X	Receiver Audio In 4A
Y	Receiver Audio In 4B
k	Receiver Audio In 5A
m	Receiver Audio In 5B
F	Navigation Receiver Audio 1A
G	Navigation Receiver Audio 1B
a	Navigation Receiver Audio 2A
b	Navigation Receiver Audio 2B
n	Navigation Receiver Audio 3A
p	Navigation Receiver Audio 3B
c	Navigation Receiver Audio 4A
H	Navigation Receiver Audio 4B
J	Navigation Receiver Audio 5A
K	Navigation Receiver Audio 5B
s	Navigation Receiver Audio 6A
t	Navigation Receiver Audio 6B
N	Direct Audio (Uncontrolled) 1A
f	Direct Audio (Uncontrolled) 1B
q	Direct Audio (Uncontrolled) 2A
r	Direct Audio (Uncontrolled) 2B
L	Direct Audio (Controlled) 3A
M	Direct Audio (Controlled) 3B
d	Direct Audio (Controlled) 4A
e	Direct Audio (Controlled) 4B
Z	Remote Xmit Sel 1
T	Remote Xmit Sel 2
h	SEQUENCE/SP5T SEL
i	MASTER/SLAVE Control

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

3.11.1 Temperature-altitude (see 4.8.1). The equipment shall comply with the temperature-altitude requirements set forth in MIL-E-5400 for class 1A equipment. In addition, it shall withstand air transportation in unpressurized aircraft up to 50,000 feet above sea level. The input supply voltages shall be from 17.0 to 33.0 volts dc.

3.11.2 Humidity (see 4.8.2). The equipment shall be operable without degradation in specified performance, and shall sustain no physical damage, during and after prolonged exposure to extreme high humidity levels as encountered in tropical areas.

3.11.3 Dust (see 4.8.3). The equipment shall withstand, in both operating and nonoperating conditions, exposure to dust particles.

3.11.4 Salt fog (see 4.8.4). The equipment shall be resistant to the corrosive effects of salt-sea atmosphere.

3.11.5 Fungus (see 4.8.5). The equipment, in the assembled and ready for delivery condition, shall provide no nutrients in material, coating or contaminant for or support fungal growth when tested in accordance with 4.8.5. Only inherently fungus resistant grades of materials per requirement 4, MIL-STD-454 shall be used. Prior to delivery, all fungus nutrient contaminants or residues shall be removed.

3.11.6 Explosive atmosphere (see 4.8.6). The equipment shall not cause ignition of an ambient-explosive gaseous mixture with air when operated in such an atmosphere.

3.11.7 Rain (see 4.8.7). The equipment shall be operable without degradation in specified performance, and shall sustain no physical damage during periods of heavy precipitation with intermittent wind.

3.11.8 Acceleration (see 4.8.8). The equipment shall be operable without degradation in specified performance, and sustain no physical damage, when exposed to steady-state acceleration levels of up to 6g for a duration of 1 minute.

3.11.9 Vibration and shock.

3.11.9.1 Vibration (see 4.8.9.1). The equipment (with vibration isolators in place, if any) shall operate satisfactorily when subjected to vibration within the frequency range and amplitude as shown in Figure 2, Curve IIIb, Sheet 3 of MIL-E-5400, truncated at 500 Hz.

3.11.9.2 Shock.

3.11.9.2.1 Basic design (see 4.8.9.2.1). The equipment shall not suffer damage or subsequently fail to provide specified performance when subjected to 18 impact shocks of 15 g, consisting of three shocks in opposite directions along each of three mutually perpendicular axes. Each shock impulse shall have a time duration of 11 ± 1 millisecond. The "g" value shall be within ± 10% when measured with a 0.2 to 250 Hz filter, and maximum "g" value shall occur at approximately 5 ± 1/2 milliseconds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

3.11.9.2.2 Bench handling (see 4.8.9.2.2). The equipment shall withstand shocks encountered in servicing.

3.11.9.2.3 Crash safety (see 4.8.9.2.3). The equipment shall withstand 12 impact shocks of 30 g's, consisting of two shocks in opposite directions along each of three mutually perpendicular axes. Each shock impulse shall have a time duration of 11 ± 1 milliseconds. The "g" value shall be within ±10 percent when measured with a 0.2 to 250 Hz filter and the maximum "g" value shall occur at approximately 5 ± 1/2 milliseconds. Bending and distortion shall be permitted, however, there shall be no failure to the attaching joints and the equipment or dummy load shall remain in place. Note: A dummy mass with outside structure and mounting points identical to the flight hardware and mass-loaded to be dynamically the same as the flight hardware may be used for the crash safety test.

3.12 Voltage protection, low and reverse (see 4.8.11). The unit(s) shall be subjected to and meet the requirements of 3.13.1 and shall show no visual damage after being subjected to 33 volts dc reverse polarity and continuous operation at 17 to 33 volts dc.

3.13 Electrical requirements (see 4.10).

3.13.1 Input power (see 4.10.1). The unit(s) shall operate from normal 27.5 volt dc aircraft power as defined by MIL-STD-704, and shall maintain required performance over the input voltage range of 22.0 to 33.0 volts dc. It shall also maintain required performance when subjected to input overvoltage. During emergency mode, 17.0 volts to 22.0 volts, and undervoltage conditions, the unit will operate with degraded performance. When the input current is measured with 29.0 volt dc input, 1.25 volts audio output at the headset pins (before transmit attenuation (see 3.13.20)) and 0 dBm output on one transmitter audio output line, the CSC input current shall not exceed 250 milliamps, 7.25 watts. The input power requirements for front panel display lamps shall be no greater than:

a. 28 volts dc, 4.0 watts (red), 6.5 watts (green)

b. 5 volt ac/dc, 4.0 watts (red), 6.5 watts (green)

c. 115V ac, 400 Hz, 3 watts

3.13.2 Output interphone (see 4.10.2).

3.13.2.1 ICS line. With the selector switch set in any position PVT, ICS or 1 through 5, a 1.0 kHZ signal at 0.6 volts applied to the required pad (see 4.7.f), which is connected to pins J1-L and J1-M (microphone input) and pin J1-b jumpered to ground, the output across pins J2-R and J2-S shall be between .35 and .43 volts; this output shall also be the same for any remote channel selection and also when the HOT MIC-VOX ON-NORM-OFF switch is set to HOT-MIC or VOX ON with VOX control fully counterclockwise and pin J1-b is ungrounded. HOT MIC and Interphone key (pin J1-b) operation shall be overridden when any radio channel is selected and keyed. HOT MIC and VOX ON shall be overridden when PVT is selected and J1-A or J1-d is jumpered to ground.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:19Z
Check the source to verify that this is the current version before use.

3.13.2.2 PVT line. With the selector switch set in PVT position, a 1.0 kHz signal at 0.6 volts applied to the required pad (see 4.7.f), to pins J1-L and J1-M (microphone input), the output across pins J2-g and J2-P shall be between .35 and .43 volts when pin J1-A is jumpered to ground (J1-c). Pin JI-A may be grounded in the aircraft wiring to provide automatic HOT MIC activation when selector switch is set to PVT.

3.13.2.3 ICS off. Any ICS signal to the headset shall be reduced at least 60 dB when switch is set to ICS OFF position.

3.13.3 Sidetone (see 4.10.3). With selector switch set to ICS position, HOT MIC-VOX ON-NORM-ICS OFF switch to NORM position, pin J1-d or J1-b jumpered to ground and 1.0 kHz signal at 0.6 volts applied to the required pad, the output voltage across pins J2-D and J2-E shall be -4.0 to -6.0 dB below the level obtained across J2-D and J2-E when a 0.39 volt rms, 1.0 kHz signal is applied to pins J2-R and J2-S. A comparable sidetone reduction shall occur when selector switch is set to PVT and pin J1-A is jumpered to ground.

3.13.4 Response, microphone amp (see 4.10.4). The frequency response shall not vary more than ±1.0 dB within the freq range of 700 Hz to 2250 Hz. The 3 dB ±1 dB bandwidth shall be 350 Hz to 4500 Hz. The 1 dB and 3 dB ± 1 dB levels are referenced to the level at 1kHz. The amplitude between 350 Hz and 4500 Hz shall be no greater than 1 dB above the amplitude at 1 kHz. The slope of the asymptotic attenuation curve above 4500 Hz and below 350 Hz shall be no less than 6 dB per octave.

3.13.4.1 Response, wideband audio out (see 4.10.4.1). The frequency response shall not vary more than ±1.0 dB in the range 200 to 3000 Hz. The 3 dB ± 1 dB bandwidth shall be 100 to 6000 Hz. These levels are referenced to the 1 kHz level. The slope of the asympotic attenuation curve above 6000 and below 100 should be no less than 6 dB per octave.

3.13.5 Distortion, microphone amp (see 4.10.5). The total harmonic distortion shall not exceed three (3) percent with a signal level of 0.6 volts applied to the required pad and five (5) percent with a signal level of 4.0 volts applied to the required pad over the frequency range of 350 to 4500 Hz. Center clipping ("zero-crossing distortion") shall be no greater than 2 dB with a signal level of 0.085 volts at 1.0 kHz applied to the required pad.

3.13.6 Attack and release time, AGC (see 4.10.6). With 1.0 kHz applied to pins J1-L and J1-M, the attack time shall be not more than 0.1 seconds and the release time shall be 3.0 seconds ±1.5 seconds.

3.13.6.1 Attack and Release Time VOX (see 4.10.6.1). With 1.0 kHz applied to pins J1-L and J1-M, the attack time shall be not more than .1 seconds and the release time shall be between .5 and 1.5 seconds.

3.13.7 Breakpoint and slope, AGC (see 4.10.7). With 1.0 kHz applied to the pad connected to pins J1-L and J1-M (see 4.7.f) and when the linearity of the

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:19Z
Check the source to verify that this is the currentversion before use.

input-output curve varies 1.0 dB (this will be the breakpoint), the input signal level shall be between 0.2 and 0.5 volts. With an increase in the input signal level from 0.5 to 5.0 volts, the slope of the input-output curve from the breakpoint shall be 2 dB maximum change in output for a 20 dB change in input. This requirement applies to the equipment operated in both the 5 ohm and 150 ohm microphone input positions.

3.13.7.1 Sensitivity of VOX (see 4.10.7.1). The sensitivity of the VOX circuit shall be at least 8 dB greater at 500 Hz than at 3 kHz. At the maximum counterclockwise position of the VOX control and with the ICS switch at VOX ON the unit will operate exactly as it does for the ICS switch at HOT MIC.

3.13.8 Transmit audio output, microphone amp (see 4.10.8). With J2-h open and J1-d connected to J1-c (key common) or with J2-h connected to key common (manual SP5T mode) and any of J2-Z, T or JI-a, F, B, d connected to key common, the transmit audio shall be enabled as follows: With the microphone selector switch in position 1 and the channel selector switch in positions 1 through 5 and a 1.0 kHz signal at a level of 0.6 volts applied to the 5000 to 5 ohm pad connected to the microphone input, the output on the indicated output lines shall be 0.39 volts ±10%. With the microphone selector switch in position 2 and a 1.0 kHz signal at a level of 0.6 volts applied to a 2.4K to 150 ohm pad with blocking capacitor, 47 µfd or greater, connected to the microphone input, the output on the indicated output lines shall be 0.39 volts ±10%. The wideband audio output J1-K shall be 28 mV ±10% for either position of the MIC switch.

3.13.8.1 Aux Mic Input (see 4.10.8.1). With an input of 28mV at 1 kHz the output on the indicated output lines shall be 0.39V ±10%.

3.13.8.2 Remote Transmit Selection (see 4.10.8.2). With the transmit select control operated to RMT the operation shall be as follows:

3.13.8.2.1 Sequential Step Mode (see 4.10.8.2.1). With power on and +28 volts supplied through a resistor to limit the current to !ess than 1 Ampere to each of pins J1-f, g, h, r, t, transmit selection shall be accomplished sequentially, one channel at a time, by momentarily connecting pin J1-B to J1-c. Transmit selection shall be in the sequence 1, 2, 3, 4, 5, 1, 2, etc. Performance of the unit shall meet the same requirements as for manual selection of any channel. With +28 volts removed from J1-g, channel 2 shall be skipped and so on. If all above pins are disconnected from +28 volts, channel 1 shall be selected. At power turn-on the first channel with +28 volts on it will be selected.

3.13.8.2.2 SP5T Switch Mode (see 4.10.8.2). With pin J2-h connected to J1-c (key common), the unit shall be in receive mode unless one of the following pin connections is made.

3.13.8.2.2.1 With pin J2-Z connected to J1-c, channel 1 shall be selected and keyed. Performance of the unit shall meet the same requirements as for manual selection of channel 1 with J1-d connected to J1-c (Radio PTT ON).

3.13.8.2.2.2 With pin J2-T connected to J1-c, channel 2 shall be selected and keyed. Performance of the unit shall meet the same requirements as for manual selection of channel 2 with J1-d connected to J1-c (Radio PTT ON).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

3.13.8.2.2.3 With pin J1-a connected to J1-c, channel 3 shall be selected and keyed. Performance of the unit shall meet the same requirements as for manual selection of channel 3 with J1-d connected to J1-c (Radio PTT ON).

3.13.8.2.2.4 With pin J1-F connected to J1-c, channel 4 shall be selected and keyed. Performance of the unit shall meet the same requirements as for manual selection of channel 4 with J1-d connected to J1-c (Radio PTT ON).

3.13.8.2.2.5 With pin J1-B connected to J1-c, channel 5 shall be selected and keyed. Performance of the unit shall meet the same requirements as for manual selection of channel 5 with J1-d connected to J1-c (Radio PTT ON).

3.13.8.3 Piggy Back Operation (see 4.10.8.3). Two units may be operated in tandem to provide capability for 10 radios, 12 NAV channels, four DIRECT AUDIO (two uncontrolled and two controlled) channels and two PVT channels and two ICS channels. Two units shall be configured for piggy back operations by external connections only. One of the two units will be selected for master and one for slave status. Master status shall be selected by externally connecting J1-a to J1-c (key common). Slave status shall be selected by connecting J2-T to J1-c and leaving J1-a open. Remote operation in a piggy-back configuration shall be by sequential step mode only.

When two units are configured for piggy-back operation it shall be possible to operate with both units in manual, both units in remote or one unit in manual and the other in remote.

3.13.8.3.1 Interphone operation. When the ICS is keyed by ICS PTT (J1-b to J1-c) or master unit HOT MIC or by master unit VOX or by RAD PTT with ICS selected on master unit, the master unit ICS audio circuit shall be selected and keyed unless the master unit is in remote and the slave unit is in manual with ICS selected.

3.13.8.3.2 Master unit in manual mode. When the master unit is in manual mode, the two unit station shall meet all receive mode requirements for all receive channels. Channel selection shall be by the select switch of the master unit only.

3.13.8.3.3 Slave unit in manual mode. When the slave unit is in manual mode and the master unit is in manual mode, operation shall be in accordance with 3.13.8.3.2. When the slave unit is in manual mode and the master unit in remote mode, channel selection will be only by means of the slave unit select control. When a transmit channel is not being keyed the station will meet all the receive mode requirements for all receive channels. When either PVT or ICS is selected on the slave unit mic audio will be applied to the selected interphone line corresponding to the slave unit. When the PVT channel or a transmit channel is selected but not keyed and interphone is keyed mic audio will be applied to the ICS line corresponding to the master unit.

3.13.8.3.4 Both units in remote mode. When RMT is selected on both units transmit selection will be by sequential selection only (momentary ground to J1-B). Channel selection will be per paragraph 3.13.8.2.1, the next channel after channel 5 on any unit will be channel 1 on the other unit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

3.13.9 Audio output, headset amp (see 4.10.9). The headset amp shall contain an internally adjustable control to permit adjustment of the output level of the headset amp. With an input of 2.75V, 1 kHz at any one radio, NAV and DIRECT channel or an input of 0.39V, 1 kHz at either the ICS or PVT interphone input, the output across 8 ohms shall be adjustable from below 1.0 volts to above 1.6 volts. With the MIC switch in position 2 (150 ohms) after the output level was previously set to 1.25 volts with MIC switch in position 1, the output level shall be 1.25V ± 10%.

With the control set for 1.25 volts output across 8 ohms the amplifier output voltage shall be limited when the input voltage to a radio NAV or DIRECT channel cross a threshold between 2.75 and 3.87 volts. The limiter circuitry shall operate effectively through 5.47 volts of input, to limit the output to within 4 dB of the maximum set value of 1.25V. The output level of a received tone shall not be reduced by more than 9 dB by the presence of another tone greater than the limiting level at any RADIO, NAV, or DIRECT channel.

3.13.10 Response, headset amp (see 4.10.10). The frequency response shall not vary more than ±1.0 dB within the freq range of 700 Hz to 2250 Hz. The 3 dB ± 1 dB bandwidth shall be 350-Hz to 4500 Hz. The 1 dB and 3 dB levels are referenced to the level at 1 kHz. The amplitude between 350 Hz and 4500 Hz shall not be greater than 1 dB above the amplitude at 1 kHz. The slope of the asymptotic attenuation curve above 4500 Hz and below 350 Hz shall be no less than 6 dB per octave.

3.13.11 Distortion, headset amp (see 4.10.11). When frequencies over the range of 350 to 4500 Hz are applied with the required input voltages to audio input lines, the total harmonic distortion at the headset amp output shall not exceed ten (10) percent. Center clipping ("zero-crossing distortion") shall be no greater than 2 dB below the peak output level with a signal level of 0.39 volts at 1.0 kHz applied to the receive audio input lines or .055 volts applied to the interphone lines.

3.13.12 Broadband Noise, Headset (see 4.10.12). The headset broadband noise output of the unit(s) shall not exceed 0.9 millivolts when subjected to the test of 4.10.12.

3.13.13 Isolation, monitor switches/audio lines (see 4.10.13). With 1.0 kHz applied to the specified audio inputs and a 1.25V output at the headset amplifier, the output shall not exceed 1.25 mv when the monitor switch is set to the "Off" position and the selector switch is set to any other audio input line. With 2.75 volts, 1.0 kHz applied to any audio input line, monitor switches to "ON" position, the audio level on any other audio line shall be less than 2.75 millivolts. (When the input signal is applied to one audio input, the signal shall be less than 27.5 millivolts on the two common NAV audio lines or the common DIRECT line.)

3.13.14 Isolation, transmitter lines (see 4.10.14). With 2.75 volts, 1.0 kHz applied to any monitor input line or 0.39 volts, 1.0 kHz applied to the ICS or PVT lines and the headset amplifier output set for 1.25V, the undesired voltage level at each transmitter audio output or control line shall be at least 120 dB below the normal 0.39 volts output, except that on the selected transmitter audio output line the undesired signal shall not be detectable in the output noise floor as observed

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

using a 5 Hz bandwidth. The output noise floor as observed using a 5 Hz bandwidth shall be no greater than -90 dB referenced to 0.39V out. Similarly, a 1.0 kHz, 0.39 volt, signal on any selected ICS, PVT or transmit audio line shall have an isolation of at least 120 dB below normal, 0.39 volts output at any unselected transmit audio line.

3.13.15 Transient protection (see 4.10.15). The unit(s) shall meet specified requirements after subjection to a +80 volt pulse and a -80 volt pulse for 0.2 second duration three (3) times within 1 minute. In addition, the unit shall meet specified requirements after subjection to a +600-volt spike of 5-microseconds ±0.2 microseconds duration, applied three (3) times within 1 minute.

3.13.16 Headset amp, unterminated (see 4.10.16). The headset amp output shall be stable (no oscillation) when pins J2-D and J2-E are open circuited, or short circuited and grounded, and the monitor and selector switches are placed to any position and all volume controls are rotated to any level between minimum and maximum audio output.

3.13.17 Transmitter control (see 4.10.16). With any of radio channels 1 thru 5 selected, pin J1-d jumpered to ground, and a dc voltage applied to the specified pins, the transmitter control lines shall withstand 1.0 amps through the lines without any degradation.

3.13.18 Function selector logic (see 4.10.18). With the selection of any one of PVT, ICS, 1,2,3,4, or 5, a separate .5-ampere capacity circuit pin to ground shall be provided. The normally open pin shall be grounded (dc Power Gnd) when its respective function is selected.

3.13.19 Microphone power (see 4.10.19). The unit shall provide up to 10 mA at 10-volts dc ±20% on the microphone high input (J1-L) with power return through the microphone low input (J1-M) when the microphone selection switch is set to position 2.

3.13.20 Received audio input attenuation during transmit (see 4.10.20). With a 0.39 volt rms, 1.0 kHz input applied to PVT or ICS input or a 2.75 volt rms, 1.0 kHz input applied to any other audio input and the headset amplifier output set for 1.25V rms, the received audio inputs appearing at the headset amplifier output shall be selectively attenuated 12 dB ± 1 dB (below the 1.25Vrms level) as shown in table II when selector switch is set to positions 1 through 5 and pin J1-d is grounded. With J1-F and J1-d both grounded to key common, all the received audio inputs except DIRECT channels shall be attenuated 12 dB ± 1 dB.

3.13.21 Audio level (volume) adjustments (see 4.10.21).

3.13.21.1 Main volume control. With all volume controls set at maximum audio output and the input to a specific audio input line adjusted to obtain 1.25 volts output at the headset, the output shall be reduced by 40 ± 3 dB when the main volume control is set for minimum audio output.

3.13.21.2 Monitor audio level adjustment. With all volume controls set at maximum audio output and the input to a specific audio input line adjusted to obtain 1.25 volts output at the headset, the output shall be reduced by 27 ± 2dB when the individual audio level control (for the selected radio audio monitor

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

Table II. Received audio input attenuation

SELECTOR SWITCH POSITION	PVT	ICS	RECEIVED AUDIO LEVEL					NAV 1	NAV 2	DIRECT RADIOS
			1	2	3	4	5			
1	-12 dB	-12 dB	0 dB	-12 dB					-12 dB	0 dB
2	-12 dB		-12 dB	0 dB	-12 dB				-12 dB	0 dB
3	-12 dB			-12 dB	0 dB	-12 dB			-12 dB	0 dB
4	-12 dB				-12 dB	0 dB	-12 dB		-12 dB	0 dB
5	-12 dB					-12 dB	0 dB	-12 dB	-12 dB	0 dB
PVT	Sidetone	0 dB								0 dB
ICS	0 dB	Sidetone	0 dB							0 dB

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

input) is set for minimum audio output. The audio outputs of all other monitor lines shall be unaffected by this adjustment. This requirement shall be met for any headset amplifier adjustment level between 1.1 VRMS and 1.6 VRMS.

3.13.22 TEMPEST. The equipment shall comply with the requirements of Confidential Appendix A, titled "TEMPEST Requirements" for the Control Unit, Communication System C-11746(V)( )/ARC.

3.14 Electromagnetic interference (see 4.11). The communications system control units shall comply with the requirements of Table 1 of MIL-STD-461A Notice 4 as defined herein:

CE01	CS01	RE02	RS02
CE02	CS02	RE02.1	RS03.1
CE03	CS06		
CE04			
CE05			

3.14.1 Audio susceptibility, CS01 (see 4.11.1). The unit(s) shall comply with the CS01 requirements. The unit(s) headset (J2-D and J2-E) output shall not exceed 4.0 MV between 30 Hz and 50 KHz. The interphone (J2-R and J2-S) output shall not exceed 13 MV between 30 Hz and 50 KHz.

3.14.2 Conducted susceptibility CS02 (see 4.11.2). The requirements of CS02 shall apply except that all RF signals be modulated 50 percent AM at 1.0 kHz. The success criteria for this requirement shall be a maximum noise output for the Headset Amplifier (J2-D and J2-E) as follows: 4.0 MV between 150 kHz and 199 MHz, and between 251 MHz and 400 MHz; 4.0 MV between 200 MHz and 250 MHz. The maximum noise output for the Interphone Terminals (J2-R and J2-S) shall be 13.0 MV.

3.14.3 Radiated susceptibility RS03 (see 4.11.3). The requirement of RS03 shall apply except that all RF signals shall be modulated 50 percent AM at 1.0 kHz. The field strength for RS03 shall be as specified in MIL-STD-461 except that it shall be increased to 10 volts/meter over the range 2 to 30.0 MHz. The success criteria for this requirement shall be a maximum noise output of 4.0 MV for the Headset Amplifier (J2-D and J2-E) and 13.0 MV at the Interphone Terminals (J2-R and J2-S).

3.14.4 Radiated susceptibility induction field (see 4.11.4). The unit(s) headset noise output (J2-D and J2-E) shall not exceed 10 MV and the interphone noise output (J2-R and J2-S) shall not exceed 20 MV when subjected to the test of 4.11.4.

3.14.5 Bonds, grounds, connectors and cables (see 4.14). Bonding for the equipment current path return, RF potentials and shock hazard shall be so installed that expansion, contraction or movement incident to normal service use will not break or loosen the connection. Surface preparation for bonds and grounds shall be accomplished by removing all anodic film, grease, paint, lacquer or other high resistance properties from the immediate area of contact. Bonding impedance shall not exceed 2.5 milliohms. Case and chassis mounted connectors shall have approved RF conductive finish and shall be installed as above. Bonding impedance of the connector shell to the case shall not exceed 2.5 milliohms. Interconnecting cables

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

requiring shields shall have a minimum of 40 dB shield attenuation to electric fields over the frequency range of 10 kHz to 400 MHz. The shield shall be constructed of tinned copper braid. Connectors with cables having shields shall have an approved RF conductive finish and provided with EMI backshells for terminating the shields.

3.15 Burn-in (see 4.6). Burn-in shall be performed as described in paragraph 4.6.2.

3.16 Reliability program. The unit(s) shall be subjected to a reliability program that shall provide for, but not be limited to, the following reliability tasks as described in MIL-STD-785:

a. Reliability Management

b. Parts Reliability

c. Reliability Demonstration

d. Failure Data

3.17 Reliability (see 4.6). The equipment minimum Mean-Time-Between Failures (MTBF, Θ_1) shall be 3500 hours when used in the following manner:

Ambient Temperature: -54°C to +55°C

Duty Cycle: 100 percent (5 minutes receive, 1 minute transmit)

Environment: Airborne

This requirement shall be demonstrated in accordance with paragraph 4.6.3.

3.18 Operational inspection (see 4.13). Each unit shall be operated as specified after applicable processes and inspections are completed to determine that each unit is operable prior to packaging. Voice quality shall indicate proper operation of the unit(s).

3.19 Interchangeability (see 4.12). Like units, assemblies, subassemblies, and replaceable parts shall conform to Requirement 7 of MIL-STD-454 and shall be physically and functionally interchangeable without modification of such items or of the equipment.

3.20 Systems safety engineering (see 4.14).

3.20.1 Personnel hazards. Personnel hazards shall be kept to a minimum through compliance of configuration changes and parts selection with Requirement 1 of MIL-STD-454. Compliance with these requirements will be verified through a visual inspection.

3.20.2 Edge rounding exposed. Exposed edges shall be rounded to a minimum radius of 0.04 inch (1 mm), and exposed corners to a minimum of 0.5 inch (13 mm).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

3.20.3 Radioactive materials. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes, surge arrestors and lenses).

3.21 Workmanship (see 4.14). Workmanship shall be in accordance with Requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection and test requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection and test requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections and tests set forth in the specification where such inspections and tests are deemed necessary to assure supplies and services confonm to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First Article inspection (does not include packaging (see 4.3)).

b. Inspection covered by subsidiary documents (see 4.4).

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before packaging (see 4.5).

(2) Quality conformance inspection of packaging (see 4.15).

4.3 First article. Unless otherwise specified in the contract or purchase order, the First Article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish that number of First Article units of the complete C-11746(V)( )/ARC as specified in the contract or purchase order.

4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, burn-in/reliability specified in 4.6, and the inspections specified for Group A, Group B, and Group C (see tables III, IV and V, respectively). The inspection shall be performed in the following order: 4.4, 4.6, Group A and Group B for all units, and designated units for specific subgroups within Group C in accordance with 4.5.4.1. After completion of Group C environmental tests, conforming units shall be reinspected and shall pass all Group A inspections.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection before preparation for delivery.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:19Z
Check the source to verify that this is the currentversion before use.

Item	Where Required
Finish	3.4
Marking, Lighting and Interface	3.6
Printing Wiring Assemblies	3.8
Semi-conductors	3.9

4.5 Quality conformance inspection of equipment before packaging. Each unit of equipment shall be subjected to Burn-In (3.15) prior to inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. Failure to meet any requirement as a result of Burn-In shall be classified as a defect and recorded as such. The Government will review, and evaluate the contractor's inspection procedures and examine the contractor s inspection records. In addition, the Government at its discretion, may perform all or any part of the specified inspections, to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in table III. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined here, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subjected to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in table III. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operational inspection (4.13) of the entire lot (including the sample units) shall be performed last. Subsequent handling of the units shall be minimized to reduce possibility of damage to the units.

4.5.2 Group B inspection. This inspection, including sampling, shall conform to table IV and to the procedures for small sample inspection of MIL-STD-105 using the special inspection levels. The AQL shall be 6.5 percent defective and the inspection level shall be S4 for normal, tightened and reduced inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection

Inspection	Req. Para.	Insp. Para.	AQL	
			Major	Minor
Visual and Mechanical	3.20 3.21	4.14		
Printed Circuits Assemblies:				
Main XMIT Circuit Card Assy			0.65%	2.5%
Audio Circuit Card Assy			0.65%	2.5%
Receive Circuit Card Assy			0.65%	2.5%
Interconnect Board			0.65%	2.5%
C-11747(P)/ARC (less front panel)			1.0%	4.0%
C-11746(V)( )/ARC			1.0%	4.0%
Electrical:				
Output, interphone (to include ICS, PVT and ICS OFF)	3.13.2	4.10.2	*1.0% for the combined group	
Sidetone	3.13.3	4.10.3		
Response, microphone amp	3.13.4	4.10.4		
Wideband audio out	3.13.4.1	4.10.4.1		
Distortion, microphone amp	3.13.5	4.10.5.1		
Transmit audio outpout, microphone amp	3.13.8	4.10.8		
AUX MIC input	3.13.8.1	4.10.8.1		
Remote XMIT Selection				
Sequential Step Mode	3.13.8.2.1	4.10.8.2.1		
SP5T Switch Mode	3.13.8.2.2	4.10.8.2.2		
Audio output, headset amp	3.13.9	4.10.9.1		
Response, headset amp	3.13.10	4.10.10		
Distortion, headset amp	3.13.11	4.10.11.1		
Transmitter control	3.13.17	4.10.17.2		
Function Selector Logic	3.13.18	4.10.18.1		
Microphone Power	3.13.19	4.10.19		
Received Audio Input attenuation during transmit	3.13.20	4.10.20		
Audio Level (Volume) adjustments	3.13.21	4.10.21		
Operational inspection (with Test Fixture)	3.18	4.13.1	1.0%	

*All electrical defects are considered major

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

TABLE IV. Group B inspection

Inspection	Req. Para.	Insp. Para.	AQL
Subgroup I:			
Input power	3.13.1	4.10.1	6.5%
Distortion, microphone amp	3.13.5	4.10.5.2	group
Attack and release time, AGC	3.13.6	4.10.6	
Attack and release time, VOX	3.13.6.1	4.10.6.1	
Breakpoint and slope, AGC	3.13.7	4.10.7	
Sensitivity of VOX	3.13.7.1	4.10.7.1	
Piggyback controls	3.13.8.3	4.10.8.3	
Audio output headset amp	3.13.9	4.10.9.2	
Distortion headset amp	3.13.11	4.10.11.2	
Broadband noise, headset	3.13.12	4.10.12	
Isolation, monitor switches/ audio lines	3.13.13	4.10.13	
Isolation, transmitter lines	3.13.14	4.10.14	
Transient protection	3.13.15	4.10.15	
Headset amp, unterminated	3.13.16	4.10.16	
Transmitter control	3.13.17	4.10.17.1	
Function selector logic	3.13.18	4.10.18.1	
Subgroup II:			
Interchangeability (Mechanical)	3.19	4.12	6.5%
*Interchangeability (Electrical)	3.19	4.12	6.5%
Operational Inspection (system simulation)	3.18	4.13.2	6.5%

*Inspection shall be performed prior to assembly into the units.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

4.5.2.1 Subgroup I. All units selected for group B tests will be subjected to subgroup I tests.

4.5.2.2 Subgroup II. Four units of these selected for group B tests will be subjected to subgroup II tests.

4.5.3 Order of inspection within Group B. Group B inspection shall be performed in any order which is satisfactory to the Government.

4.5.4 Group C inspection. This inspection shall be as listed in table V and shall be performed on lots that have been subjected to and met Group A and Group B inspections. No defects shall be allowed. Sample unit(s) shall be selected in accordance with 4.5.4.1.

4.5.4.1 Sampling for inspection. Units shall be selected at random for each group of inspections in table V as specified in 4.5.4.1.2. The first samples selected shall be from the first quality conformance inspection production lot(s). Selection of these samples shall be as specified and within 30 calendar days (45 calendar days for the fungus test). After selection, the samples shall have been subjected to and completed the Group C inspections (see 6.3).

4.5.4.1.1 Subgroup I. For this subgroup, two (2) units from the first eight (8) production units of the first 500 or part thereof on order shall be selected for the first Group C inspection. If quantities in excess of 500 are on order, two (2) units from each successive 500 shall be selected for Group C inspection.

4.5.4.1. Subgroup II. For this subgroup, two (2) units from the first eight (8) production units shall be selected for the first Group C inspection. If quantities in excess of 1000 are on order, two (2) units from each successive 1000 shall be selected for Group C inspection. These samples may be the same units selected for Subgroup I tests.

4.5.4.1.3 Subgroup III. For this subgroup, one (1) unit from the first eight (8) production units of the first 1000 or part thereof on order shall be selected for the first Group C inspection. If quantities in excess of 1000 are on order, one (1) unit from each successive 1000 shall be selected for Group C inspection.

4.5.4.2 Order Order of inspection within Group C. Group C inspection shall be performed in an order which is satisfactory to the Government.

4.5.4.3 Group C failures. Actions required relative to Group C failures shall be as specified in the contract.

4.5.5 Reinspection of conforming Group C sample units. Unless otherwise specified, lots which have been subjected to and passed Group C inspection may be delivered on the contract or order, provided all damage is repaired, cleaned of all foreign matter, and the sample units are resubjected to and pass Group A inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

TABLE V. Group C inspection

	REQ. PARAGRAPH				INSPECTION PARAGRAPH
	UNIT 1	UNIT 2	UNIT 3	UNIT 4	
Subgroup I:					
Weight		3.2.1			4.14.1
Voltage protection		3.12			4.9.1
Temperature-altitude	3.11.1				4.8.1
Humidity	3.11.2				4.8.2
Vibration			3.11.9.1		4.8.9.2
Shock			3.11.9.2.1 & 3.11.9.2.2		4.8.9.3
Dust			3.11.3		4.8.3
Subgroup II:					
Electromagnetic interference		3.14			4.11
Explosive atmosphere		3.11.6			4.8.6
Acceleration			3.11.8		4.8.8
Fungus			3.11.5		4.8.5
Salt Fog		3.11.4			4.8.4
Rain	3.11.7				4.8.7
Voltage - Front Panel (Green)		3.6.4			4.9.2
Voltage - Front Panel (Red)		3.6.5			4.9.3
Crash Safety			3.11.9.2.3		4.8.9.2.3
Subgroup III:					
TEMPEST				3.13.22	4.10.22

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

4.6 Burn-in and reliability.

4.6.1 Conditions precedent to burn-in and reliability testing. Prior to doing any burn-in or reliability acceptance testing, the contractor shall perform a thermal and vibration survey in accordance with the paragraphs titled "Thermal Survey" and "Vibration Survey" of MIL-STD-781.

4.6.2 Burn-in. Each deliverable equipment shall be subjected to a 48-hour on-time burn-in period, the last 24 hours of which shall be failure free. Each unit which fails during the last 24 hour period shall be repaired and returned to test until it successfully survives a 24 hour period without failure. Failures which occur during burn-in test shall be noted and reported, but will not be counted towards the establishment of the equipment MTBF. Duty cycle imposed, vibration, environmental test profile and parameters to be measured shall be those included in the contractor's approved reliability test procedure.

4.6.3 Reliability testing.

4.6.3.1 The system reliability shall be demonstrated in accordance with Test Plan XVIIC, MIL-STD-781 on a minimum sample of two equipments (see Table VI). For this test plan, the MTBF (θ_1) shall be 3500 hours.

4.6.3.2 Reliability qualification test conditions. The reliability qualification test conditions shall be in accordance with Appendix B, the paragraphs titled "Combined Environments for Turboprop Aircraft" and "Helicopter Equipment" of MIL-STD-781. A failure is defined as the inability of the equipment to perform its required function. This includes catastrophic failures as well as any event which causes any departure from performance as required by this specification. Proper instrumentation shall be provided to insure immediate recognition of failure. Voltage cycling shall be according to the paragraph titled "Electrical Stress", Appendix B.

4.7 Standard test conditions. Unless otherwise specified herein, inspections shall be performed under test conditions of GENERAL REQUIREMENTS, Test Conditions, of MIL-STD-810 and the following conditions:

a. Input power: 27.5 ± 0.5-volts dc.

b. Operational duty cycle: One (1) minute talk, five (5) minute listen.

c. Warm-up time: Not more than 2 minutes.

d. Output loads: Headset amp; 8 ohms. Transmit audio output; 150 ohms (see 4.7g).

e. Input loads: Audio lines and interphone; 150 ohms (see 4.7g).

f. Input load microphone; MIC switch in position 1, 5,000 to 5 ohm pad; MIC switch in position 2, 2.4K to 150 ohm pad with blocking capacitor, 47 µfd or greater (see 4.7g).

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:19Z
Check the source to verify that this is the current version before use.

TABLE VI. Test sequence for limited sample size

UNITS AVAILABLE

3			2		1
Unit 1	Unit 2	Unit 3	Unit 1	Unit 2	Unit 1
Temperature-Alt Rain	Explosion	Dust	Temperature-Alt Rain	Dust Explosion	Temperature-Alt Dust Explosion Rain
---	---	---	---	---	---
Humidity	Salt fog	Acceleration Vibration Shock Fungus	Humidity Salt fog	Acceleration Vibration Shock Fungus	Humidity Fungus (2) Salt fog (2) Acceleration Vibration Shock

NOTES:

(1) The tests above the broken line are considered non-damaging. The order of these tests and choice of units to use may be varied if convenient.

(2) Following fungus and salt fog testing, the test item shall be thoroughly washed, cleaned, dried, and refurbished, if necessary, before proceeding with subsequent tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

g. Connector (J1) Pins: Pin pairs N-P, S-T, V-W, k-s, p-q are each terminated with a 150-ohm resistor. When MIC switch is in position 1, pins L and M should be terminated with a 5K to 5-ohm resistor pad with the junction of the 5K and 5-ohm resistors connected to pin L. When MIC switch is in position 2, Pins L and M should be terminated with 2.4K and 150 ohm pad with blocking capacitor, 47 µfd or greater, with the junction of the 2.4K and 150-ohm resistors connected to Pin L through the blocking capacitor. Pins b, c, and d, are connected to a single pole, double-throw switch, with pin c connected to the center pole. All shields are connected to pin H (chassis ground). Pin J1-K terminated with 600 ohms to power ground.

h. Connector (J2) Pins: Pin pairs A-B, V-W, U-j, X-Y, k-m, s-t, F-G, a-b, n-p, c-H, J-K, d-e, L-M, q-r, N-f, R-S, g-P are each terminated with a 150-ohm resistor. Pin pair D and E are terminated with an 8-ohm resistor. Shields should be tied to pin C.

4.8 Environmental test procedures. The unit(s) shall be subjected to the following test methods of MIL-STD-810, and other tests as specified to determine compliance with service condition requirements of 3.11. Electrical tests performed before, during and after environmental testing shall include those tests required to determine compliance with paragraphs 3.13.2, 3.13.4, 3.13.5, 3.13.8, 3.13.9, 3.13.10, 3.13.11, 3.13.16, 3.13.17, and 3.13.18.

4.8.1 Temperature-altitude. The unit(s) shall be subjected to the test of Method 504.1, Procedure 1, for equipment Category 4. In addition, the procedure shall include a 1-hour exposure (non-operation) to a pressure of 3.4 inch of Hg, simulating air-transport at 50,000 feet.

4.8.2 Humidity. The unit(s) shall be subjected to the test of Method 507.1, Procedure II for compliance with 3.11.2. In Step 6 of Procedure II, measurements shall be taken during the last five hours of the last cycle.

4.8.3 Dust. The unit(s) shall be subjected to the test of Method 510.1, Procedure 1, with the exception that the air velocity in Steps 1, 2 and 3 shall be 200 ± 100 FPM. Step 3 may proceed immediately after stabilization in Step 2 if desired.

4.8.4 Salt fog. The equipment shall be subjected to the test of Method 509.1 of MIL-STD-810. Electrical operation shall not be required. All external knobs and controls shall be operated after the test. There shall be no sticking or binding of any part, and there shall be no corrosion of finishes and metals. Such corrosion shall be defined as any visible degradation of the test item surfaces that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

4.8.5 Fungus. The equipment model in the assembled and as ready for delivery condition, shall be subjected to 28 days of Method 508.1, MIL-STD-810, except that the cyclic conditions listed under the paragraph titled "Incubation of Test Items" shall not be used. Instead, the conditions of 29°C (84°F) and 85% RH should be maintained throughout the test. The equipment test model shall not be specially cleaned for the fungus test, except for the cleaning it receives during or after production. If it is necessary for the manufacturer after assembly to remove

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

accumulated production and handling contaminants by cleaning prior to packaging and delivery, the equipment shall not be tested for three days after cleaning to allow for complete evaporation of the cleaning compound. All enclosed or gasketed assemblies shall be opened and the interior sprayed with the specified mixed spore suspension. After 14 and 28 days of test, the control items per the paragraph titled "Control Items", of Method 508.1, MIL-STD-810 shall show profuse growth over at least 50% of the area of the control items. After 28 days, based upon visual examination, the equipment assembly shall pass if it shows no more than only sparse microbial growth with restricted tuberal growth development in an area of 1 to 10% or less of the total areas and no more than six unrelated minute colonies with mycelial development in areas only in other than critical circuit portions; such as terminal spacings, printed circuit boards, etc., with sparse growth due to random contamination or traces of unmixed material ingredients. The equipment shall be operated during the inspections step (the paragraph titled "Inspection", Method 508.1, MIL-STD-810). The equipment shall fail if it does not meet operational performance criteria or if it shows more than the growth specified above.

4.8.6 Explosive atmosphere. The equipment shall be subjected to the test of Method 511.1, Procedure I of MIL-STD-810.

4.8.7 Rain. The equipment with cables in place shall be exposed to a "steady state" rainfall of 4 ± 1 inches per hour on all surfaces of the test items for a total time duration of not less than 2 hours. After completion of the rain test, the units shall be removed from the test chamber and turned on within 10 minutes after test completion. The units shall be operated and tested to determine satisfactory system operation.

4.8.8 Acceleration. The equipment shall be subjected to the test of Method 513.2, Procedure II, of MIL-STD-810. The test level shall be 6g for at least 1 minute.

4.8.9 Vibration and shock.

4.8.9.1 Vibration. The equipment when normally mounted (with vibration isolators in place, if any), shall be subjected to the test of Method 514.2, Procedure I, Category C, curve M, of Figure 514.2-3 (Part 1), with exception of the frequency range, which shall be limited to 5-500-5 Hz. The time schedules shall be in accordance with Table 514.2-III, except that the sweep time corresponding to Curve M shall be 30 minutes. The equipment shall be operated during the test. Also prior to, and following these tests, the equipment shall be operated in accordance with MIL-STD-810, General Requirements, "Performance of Test". The equipment shall not have any degradation in performance and shall not suffer any mechanical damage.

4.8.9.2 Shock.

4.8.9.2.1 Basic Design. The equipment shall be subjected to the test of Method 516.2, Procedure I of MIL-STD-810. The shock pulses shall be in accordance with Figure 516.2-2 for flight vehicle equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

4.8.9.2.2 Bench handling. The equipment shall be subjected to the test of Method 516.2, Procedure V, of MIL-STD-810.

4.8.9.2.3 Crash safety. The equipment shall be subjected to the test of Method 516.2, Procedure III, figure 516.2.2 of MIL-STD-810, for half-sine 30g, 11 ms curve, and meet the requirements of paragraph 3.11.9.2.3.

4.8.10 Test sequence. To eliminate cumulative effects of potentially damage inducing tests on a single test item, separate samples should be used for each test whenever possible. The test sequence of Table VI shall be used for limited sample size.

4.9. Power.

4.9.1 Voltage protection low and reverse. Apply plus 17 and 33 volts dc to pins J1-C and J1-D for 5 minutes at each voltage, and then apply 33 volts dc with plus to pin J1-D and negative to pin J1-C for 1 minute. The unit(s) shall meet the requirement of 3.12.

4.9.2 Front panel (green). With an adjustable 115 volt, 400 Hz ac power source applied to pins J1-J and J1-U, or an adjustable 28V dc applied to pins J1-E and J1-U or an adjustable 5V dc/ac (400 Hz) applied to J1-R and J1-U, as applicable, the unit(s) shall meet the requirements of 3.6.4.

4.9.3 Alternate front panel (red). With an adjustable 28 volts dc or an adjustable 5 volts dc/ac (400 Hz) power source applied to pins J1-E and J1-U or pins J1-R and J1-U, as applicable, the unit(s) shall meet the requirements of 3.6.5.

4.10 Electrical test.

4.10.1 Input power. With the unit(s) operating under standard test conditions except 29.0 Vdc shall be applied to Pins J1-C and J1-D. Adjust the input signal level at pins J2-R and J2-S for 1.25 volts across J2-D and J2-E. Jumper Pin J1-d to ground and apply 0.6 volts at 1.0 kHz to a 5000-ohm to 5-ohm pad connected to Pins J1-L and J1-M. While these inputs are applied, measure the dc current to determine compliance with 3.13.1. Reduce the input voltage to 21 Vdc at pins J1-C and J1-D. Verify that the performance requirements of paragraphs 3.13.2, 3.13.5, 3.13.8, 3.13.9, 3.13.11, and 3.13.18 are met. Raise the input voltage to 33 Vdc at pins J1-C and J1-D. Verify that the performance requirements of paragraphs 3.13.2, 3.13.5, 3.13.8, 3.13.9, 3.13.11, and 3.13.18 are met. Connect 28 Vdc to pins J1-E and J1-U or 5V/400 Hz to J1-R and J1-U or 115V/400 Hz to J1-J and J1-U, as applicable, and measure the input current to determine compliance with 3.13.1.

4.10.2 Output, interphone.

4.10.2.1 ICS. Set MIC switch in position 1 (5 ohms). Apply 0.6 volts at 1.0 kHz to a 5,000 to 5 ohm pad connected to pin J1-L, M and set selector switch to ICS.

4.10.2.1.1 RADIO PTT. Jumper pin J1-d to ground (J1-c) and measure the output voltage at J2-R, S to verify ICS level per 3.13.2.1.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:19Z
Check the source to verify that this is the current version before use.

4.10.2.1.2 ICS PTT. Switch jumper to J1-b to ground and measure output voltage for positions PVT, ICS and positions 1 through 5 of the selector switch. Set selector switch at RMT and momentarily jumper pin J1-B to ground to select Xmit channels 1 through 5 continuing to measure output voltage at J2-R, S to verify ICS level per 3.13.2.1. Jumper pin J2-h to ground and measure output voltage at J2-R,S to verify 3.13.2.1. Momentarily also ground J2-T, Z and J1-a, F, B one at a time and note output voltage drops to less than 10 MV. Remove jumper from J2-h to ground, and jumper J1-d to ground. Note that output voltage drops to less than 10 MV. Manually select channels 1 through 5 and note output also drops to less than 10 MV.

4.10.2.1.3 HOT MIC. Set HOT MIC-VOX ON-NORM-ICS OFF switch to HOT MIC and repeat tests of 4.10.2.1.2.

4.10.2.1.4 VOX ON. Set HOT MIC-VOX ON-NORM-ICS OFF switch to VOX ON and set VOX control port fully counterclockwise. Repeat tests of 4.10.2.1.3.

4.10.2.2 PVT line. Apply .6 volts at 1.0 kHz to a 5,000 ohm to 5.0 ohm pad connected to pin J1-L and J1-M and set selector switch to PVT position and MIC switch to position 1 (5 ohms). Jumper pin J1-A to ground and measure output voltage at J2-g to verify compliance with PVT level per paragraph 3.13.2.2. Repeat above for MIC and VOX ON positions of ICS switch. Return ICS switch to NORM. Remove jumper from J1-A and jumper J1-d to ground to again verify previous output voltage. Leave jumper J1-d connected to ground and also jumper J1-A to ground and rotate selector switch through positions ICS and 1 through 5 and note output voltage drops to less than 10 MV. Remove jumper from J1-d and rotate selector switch to RMT. Momentarily ground J1-B repetitively to select channels 1 through 5 again note output remains at less than 10 MV. Also jumper J2-h to ground and momentarily ground J2-Z and T and J1-a, F and B one at a time noting that output remains at less than 10 MV.

4.10.2.3 ICS off. Apply a 1.0 kHz input signal at .39 volts rms to pins J2-R and S. Rotate selector knob to ICS and verify 1.25 volts at J2-D and E. Rotate ICS switch to ICS OFF and verify 1.25 volts at J2-D and E. Rotate select knob to PVT and then through positions 1 through 5 and RMT. Verify that the voltage drops to less than 10 MV at all positions of the selector switch except ICS. These measurements shall be made to determine compliance with 4.10.2.3.

4.10.3 Sidetone. Apply 0.6 volts at 1.0 kHz to a 5000 ohm to 5 ohm pad connected to pins J1-L and J1-M and set selector switch to ICS position and MIC switch to position 1 (5 ohms). Jumper pin J1-b to ground and measure the output voltage across pins J2-D and J2-E. Move jumper to pin J1-A to ground, set selector switch to PVT position and measure the output voltage at pins J2-D and J2-E. Remove the jumper and apply 0.39 volts reference signal to pins J2-R and J2-S and measure the output voltage. These output voltages shall be compared to determine compliance with 3.13.3.

4.10.4 Response, microphone amp. Set MIC switch to position 1 (5 ohms) and selector switch to ICS position, jumper pin J1-d to ground, apply 0.1 volt at 1.0 kHz to a 5,000 ohm to 5 ohm pad connected to pins J1-L and J1-M and using the output level obtained across pins J2-R and J2-S as a reference level, measure the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

frequency response from 350 to 4,500 Hz and at 87 Hz and 18,000 Hz to determine compliance with 3.13.4. Repeat above after setting MIC switch to position 2 (150 ohms) and using a 2.4K ohm to 150 ohm pad with blocking capacitor, 47 µfd or greater instead of a 5,000 ohm to 5 ohm pad.

4.10.4.1 Response, wideband audio output. Set up unit as in 4.10.4 for Mic switch position 1 except use output across pins J1-K (HI) and c (LO) as a reference level. Measure the frequency response from 100 to 6,000 Hz and at 40 and 15,000 Hz to determine compliance with 3.13.4.1.

4.10.5 Distortion microphone amp.

4.10.5.1 Distortion microphone amp, hi voltage. Apply 4.0 volts at 350 Hz and 4.5 kHz to pins J1-L and J1-M through a 5000-ohm to 5-ohm pad and set selector switch to ICS position, place all monitor switches to OFF, master volume control to maximum audio output position and MIC switch to position 1 (5 ohms) and HOT MIC switch to HOT MIC position and measure the microphone amplifier output distortion to determine compliance with 3.13.5.

4.10.5.2 Distortion microphone amp, additional. Apply 0.6 and then 4.0 volts at 1.0 kHz to pins J1-L and J1-M through a 5000 ohm to 5 ohm pad and set selector switch to ICS position, place all monitor switches to OFF, master volume control maximum CW and MIC switch to position 1 (5 ohms) and HOT MIC switch to HOT MIC position and measure the microphone amplifier output distortion at each of the input voltages at 350 Hz, 1000 Hz and 4500 Hz to determine compliance with 3.13.5. Reduce the input voltages to .085 volts at 1.0 kHz and measure the zero crossing distortion to determine compliance with 3.13.5.

4.10.6 Attack and release time, AGC. Set selector switch to ICS position and MIC switch to position 1 (5 ohms) and HOT MIC switch to HOT MIC position and apply 1.34 volts at 1.0 kHz to a 5000 ohm to 5 ohm pad connected to pins J1-L and J1-M and measure the output voltage across pins J2-R and J2-S for a reference level. Remove the input signal for 30 seconds and reapply the input signal voltage, measure the AGC attack time required for the output voltage to reach within 1.0 dB of the reference level. Lower the input signal voltage to .85 ± 10%V, then reduce input voltage by 16 ± 1 dB and hold for 2 seconds. Measure output voltage for a reference level. Increase the input voltage by removing the 16 dB attenuation. Hold for 5 seconds and then insert the attenuation. Measure the release time required for the input voltage to reach within 1 dB of the reference voltage. These measurements shall be made to determine compliance with 3.13.6.

4.10.6.1 Attack and release times, VOX. Set selector switch to ICS position and MIC switch to position 1 (5 ohms) and ICS switch to VOX ON. Apply 1 volt from an audio generator to the 5,000 to 5 ohm pad at 1 kHz. Use one channel of a two channel oscilloscope to monitor the input to the pad. Use the other scope channel to monitor J2-R and S. Adjust the VOX control until the mic amplifier is enabled but a slight CCW rotation of the knob would disable it. Instantaneously reduce the audio input level by a 10 dB step attenuator and note that when the input voltage is reduced 10 dB the mic amplifier is disabled, and when increased back to 1v the mic amplifier is enabled as seen by the output signal at J2-R, S. Set input at 1 volt and with a 1 cm/sec. sweep speed on the scope instanteously decrease the input by 10 dB. The difference in time between the onset of the reduction in input level

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

and the switch-off of the output signal is the VOX release time. Then set input at -10 dB reference to 1v and suddenly switch to 1 volt. The difference in time between the sudden increase in input level and the switch-on of the output signal is the VOX attack time.

4.10.7 Breakpoint and slope, AGC. Set MIC switch to position 1 (5 ohms) and selector switch to ICS position and apply 0.1 volt at 1.0 kHz to a 5000 ohm to 5 ohm pad connected to pins J1-L and J1-M and measure the output voltage across pins J2-R and J2-S. Increase the input voltage to 0.2 volt thence to 3.0 volts and measure the output voltages. Compare the input-output voltages to determine the breakpoint. Utilizing the output voltage obtained with 0.5 volts input as a reference level, increase the input signal level to 5.0 volts and measure the output voltage, compare the input-output voltages to determine the slope. Switch the microphone selector switch to the 150 ohm position and repeat above with 2.4K to 150 ohm pad with blocking capacitor, 47 µfd or greater connected to pins J1-L and J1-M. The comparisons shall be made to determine compliance with 3.13.7.

4.10.7.1 Sensitivity of VOX. Set MIC switch to position 1 (5 ohms) and selector switch to ICS position. Apply 3 kHz at 1.1 volts to the 5,000 to 5 ohm pad. Set ICS switch to VOX ON and monitor ICS output (J2-R, S). Set VOX control fully CCW and observe ICS output on scope. Now rotate VOX control CW until ICS output just turns on. Adjust input to pad and reset VOX control if necessary until ICS output is obtained for 1.2 volts and is shut off for 1.0 volts in. Change input frequency to 500 Hz. Reduce input level and note ICS output should be off at .100V and on at .400V.

4.10.8 Transmit audio output, microphone amp. Apply 0.6 volts at 1.0 kHz to a 5,000 ohm to 5 ohm pad connected to pins J1-L and J1-M and set selector switch to position 1, also set MIC switch to position 1 (5 ohms). Connect Pin J1-d to ground. The output level shall be 0.39 volts ± 1 dB across pins J1-N and J1-P. Measure the output voltage at pin pairs J1-S and J1-T, J1-V and J1-W, J1-k and J1-s, and J1-p and J1-g, with selector switch set to positions 2, 3, 4, and 5. Set microphone switch to position 2, and connect a 2.4K to 150 ohm pad with blocking capacitor, 47 µfd or greater to pins J1-L and J1-M. Apply 0.6 volts at 1.0 kHz to the pad, set selector switch to position 1, and jumper pin J1-d to ground. Observe that the output across pins J1-N and J1-P is 0.39 volts ± 1dB. Measure the output voltage at pin pairs J1-S and J1-T, J1-V and J1-W, J1-k and J1-s, and J1-p and J1-g, with selector switch set to positions 2, 3, 4, and 5 to determine compliance with 3.13.8.

4.10.8.1 Aux Mic input. Set up equipment as in paragraph 4.10.8. Apply 30 MV at 1 kHz between J1-G (HI) and J1-c (LO). the output voltage at J1-N, P with the selector at position 1 and J1-d jumpered to ground shall be .39V ±10%.

4.10.8.2 Remote selection.

4.10.8.2.1 Remote select, sequential mode. Apply 0.6 volts at 1.0 kHz to a 5,000 to 5 ohm pad connected to pin J1-L and J1-M and set selector switch to RMT. Connect a resistor of 1K to 10K ohms between a +28 volt source and each of pins J1-f,g,h,r,t. Momentarily remove and then restore +28 volt power to the unit under test. Jumper J1-d to ground (J1-c). Measure audio output voltage at J1-S, T (channel 2). It shall be 0.39 volts ±1 dB. Remove jumper at J1-d and momentarily

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

jumper J1-B to ground then restore jumper to J1-d and now voltage at J1-S,T should go to less than 10 MV and voltage of J2-V,W should go to 0.39 volts ±1 dB. Repeat above procedure for pin pairs J1-k,s and J1-p,g (channels 4 and 5 and then J1-N,P (channel 1)).

4.10.8.2.1.1 Initialization. With channel select at RMT and the sequential mode selected (J2-h open), the selected channel shall be channel 1 immediately after power turn-on unless channel 1 radio is off or disconnected (28 volts removed from pin J1-f). In that case the first channel with a live radio is selected. If there are no live radios, channel 1 is selected.

4.10.8.2.2 Remote select, SP5T Mode. With same setup as paragraph 4.10.8.2.1 jumper J2-h to ground. Remove the jumper from J1-d to ground. Add a jumper from J2-Z to ground and measure J1-N, P audio output to be .39 volts ±1 dB. Move second jumper to J2-T to ground and measure voltage at J1-S, T. Move second jumper to J1-a to ground and measure J1-V, W. Move second jumper to J1-F to ground and measure J1-k,s. Move second jumper to J1-B and measure J1-p,g. Audio output to be .39 ± 10% volts for each channel.

4.10.8.3 Piggy back controls. Apply .6V at 1 kHz to the 5,000 to 5 ohm pad connected to pin J1-L and J1-M and set MIC switch to position 1 (5 ohms). Refer to table VII for front panel control settings and DC inputs to control lines and desired output levels and perform all 24 steps to determine compliance with 3.13.8.3.

4.10.9 Audio, headset amplifier (amp)

4.10.9.1 Audio output, single tone, headset amp. Set MIC switch to position 1 (5 ohms), set channel select switch to ICS. Adjust all volume controls to maximum clockwise and push all monitor switches in (On). Connect a 1.0 kHz input at 2.75 volts to pin pair J2-A,B. The audio output shall be 1.25 volts ± 5% at pin pair J2-D,E across 8 ohms. Apply a 3.9 volt and then a 5.4 volt input signal at 1.0 kHz to detenmine compliance with limiting threshold and maximum limited output requirements of 3.13.9. Reset input level until output is 1.25 volts RMS and then set MIC switch to position 2 (150 ohms). The audio output shall be 1.25 volts ± 10%. Return MIC switch to position 1 (5 ohms) and set input level to 2.75 volts. Then apply input in turn to following pin pairs:

J2-V,W	J2-X,Y	J2-F,G	J2-n,p	J2-J,K	J2-d,e	J2-q,r
J2-U,j	J2-k,m	J2-a,b	J2-c,H	J2-s,t	J2-L,M	J2-N,f

The audio output shall be 1.25 volts ± 5% for each of above inputs. Apply 0.39 volts RMS to pin pair J2-R,S to obtain same output and then set channel select switch to PVT and apply 0.39 volts RMS to pin pair J2-g,p to obtain same 1.25 volts ± 5% output.

4.10.9.2 Audio output, 2-tone test, headset amp. Apply a 5.4 volt input signal of 1.0 kHz to a selected input and a 2.75 volt, 1.5 kHz signal to a selected nonrelated input. Observe that the 1.5 kHz output signal is not reduced in amplitude by greater than 9 dB by the presence of the 1.0 kHz signal to determine compliance with 3.13.9.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:19Z
Check the source to verify that this is the current version before use.

Table VII. Piggy Back Control Test Settings

INPUT (NOTE 1)										OUTPUT								
	RAD 1	RAD 2	RAD 3	RAD 4	RAD 5	PTT RAD	PTT PVT	SEL CTRL	XMT 1	STAT PVT	STAT ICS	STAT 1	STAT 5	SER OUT	PVT AUD	ICS AUD	XMT AUD	NOTES
ITEM	J2 T	J2 Z	J1 a	J1 F	J1 B	J1 d	J1 A	-	J1 f	J1 X	J1 Y	J1 Z	J1 n	J2 i	J2 g,P	J2 R,S	J1 N,P	
1	HI	HI	LO	LO	HI	LO	HI	PVT	HI	LO	HI	HI	HI	LO	YES	NO	NO	
2						HI									NO	YES		
3		LO	HI							HI						NO		
4				HI						LO						YES		
5							LO								YES	NO		
6							HI	ICS		HI	LO				NO	YES		
7				LO							HI					NO		
8		HI	LO								LO					YES		
9								1			HI	LO						
10		LO	HI									HI				NO		
11				HI								LO				YES		
12						LO			LO							NO	YES	
13				LO					HI			HI					NO	
14			LO						LO			LO					YES	
15				HI	NOTE	HI		RMT	HI			NOTE				YES	NO	SEE NOTE 2
16		HI		LO	HI							HI		HI				
17	LO															NO		
18	HI	LO																
19	LO	HI		HI								LO		LO		YES		
20	HI					LO			LO							NO	YES	
21		LO	HI		NOTE	HI			HI							YES	NO	SEE NOTE 2
22					HI	LO			LO							NO	YES	
23				LO					HI			HI		HI			NO	
24				HI		HI						LO		NOTE		YES		SEE NOTE 3

NOTE 1: PIN J2-h, J1-b ARE HI AND ICS SW AT HOT MIC.

NOTE 2: IN THIS MODE, MOMENTARILY GROUNDING J1-B WILL SELECT NEXT CHANNEL, REPEAT UNTIL CHANNEL 1 IS SELECTED.

NOTE 3: IN THIS MODE, AFTER CHANNEL 5 IS SELECTED THE NEXT TIME J1-B IS MOMENTARILY GROUNDED THEN J2-i WILL MOMENTARILY GO HI FOR ABOUT 50 MS. AND THEN GO LO AGAIN AS CHANNEL T IS SELECTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

4.10.10 Response, headset amp. Set MIC switch to position 1 (5 ohms) and apply a 1.0 kHz input at 0.39 volts to pins J2-R and J2-S and adjust volume control for a 0.7 volt output across pins J2-D and J2-E and using this output level as a reference level while maintaining a constant input level, measure the frequency response from 350 to 4,500 Hz and 87 Hz and 18,000 Hz to determine compliance with 3.13.10.

4.10.11 Distortion, headset amp.

4.10.11.1 Distortion, 350 Hz, headset amp. With MIC switch in position 1, volume control fully CW and all monitor controls in (ON) and fully clockwise apply 0.39 volts to ICS and PVT audio inputs and 2.75 volts to each radio audio input in turn and measure the headset amp output distortion at 350 Hz for compliance with 3.13.11.

4.10.11.2 Distortion, full range, headset amp. Apply 0.39 volts to ICS and PVT audio inputs and 2.75 volts to each radio channel audio input in turn to obtain 1.25 volts ± 5% output at 1.0 kHz across pins J2-D and J2-E. Measure the headset amp output distortion at 350 Hz, 1000 Hz and 4500 Hz for compliance with 3.13.11. Reduce the input voltage to .055 volts for ICS and PVT audio inputs and to 0.39 volts for each radio channel audio input and measure the zero crossing distortion to determine compliance with 3.13.11.

4.10.12 Broadband noise, headset. Set the selector switch to ICS position and all received audio monitor switches to the ON position. With no received audio, ICS or PVT input signals, measure the broadband noise across pins J2-D and J2-E to determine compliance with 3.13.12.

4.10.13 Isolation, monitor switches/audio lines. Set selector switch to position 1, all monitor switches out (OFF) and maximum CW, HOT MIC switch to NORM and MASTER volume to maximum CW. Apply a 1.0 kHz signal at 2.75 volts to pin pair J2-A,B to obtain 1.25V ± 5% output across J2-D,E. Set selector switch to ICS position and measure the output level relative to the 1.25 volt output to determine compliance with 3.13.13. In turn, apply the input to pin pairs J2-V,W and J2-U,j and J2-X,Y and J2-k,m and J2-F,G and J2-a,b and J2-n,p and J2-c,H and J2-J,K and J2-s,t. As each input is applied, measure the output level relative to the 1.25 volt output to determine compliance with 3.13.13. Set selector switch to position 2, apply above input to J2-V,W to obtain 1.25V ± 5% output. Set selector switch to position 3, apply above input to J2-U,j to obtain 1.25V ± 5% output. Set selector switch to position 4, apply above input to J2-X,Y to obtain 1.25V ± 5% output. Set selector switch to position 5, apply above input to J2-k,m to obtain 1.25V ± 5% output. Leave selector switch in position 5 and apply 0.39 volt signal input to pin pair J2-R,S to obtain 1.25V + 5% output. Set Hot MIC switch to ICS OFF and measure the output relative to 1.25V to determine compliance with 3.13.13. Return Hot MIC switch to NORM and push all monitor switches in (ON) and apply 2.75V, 1.0 kHz to a radio input pair and measure the output level at the other radio input pin pairs. Repeat these measurements until all audio lines have been tested to determine compliance with 3.13.13.

4.10.14 Isolation, transmitter lines. Apply 0.39 volts at 1.0 kHz to pin pair J2-R and J2-S and obtain 1.25 volts at J2-D, E. Set MIC switch to position 1 (5 ohms) and selector switch to positions 1 through 5 and measure crosstalk, with and

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

without pin J1-d jumpered to ground, at pin pairs J1-N and J1-P, J1-S and J1-T, J1-V and J1-W, J1-K and J1-s, and J1-p and J1-g respectively, while applying in turn 0.39 volts to ICS and PVT inputs and 2.75 volts to all monitor inputs except that corresponding to the selected channel. Also measure at pins J1-f, J1-g, J1-h, J1-r and J1-t, terminated in 270 ohms to chassis ground, as selector switch is set to positions 1 through 5, respectively, and pin J1-d is jumpered to ground for each input.

Apply 0.6 volts at 1.0 kHz to a 5,000 ohm to 5 ohm pad connected to pins J1-L and J1-M. Set MIC switch to position 1 and selector switch to position ICS, PVT and 1 through 5 and measure crosstalk on all transmit audio pairs, except that corresponding to the selected channel, with the appropriate key line jumpered to ground. Now set MIC switch to position 2 and measure crosstalk on all unselected transnit audio pairs at each position of the selector switch while applying 0.6 volts at 1.0 kHz to a 2,400 ohm to 150 ohms pad connected to pins J1-L and J1-M with the appropriate key line jumpered to ground. The crosstalk levels at the transmitter audio and transmitter control lines shall comply with 3.13.14.

4.10.15 Transient protection. Under standard test conditions, apply a +80 volt pulse and -80 volt pulse for 0.2 second duration between pins J1-C and J1-D. Apply each pulse three (3) times within one (1) minute as specified in MIL-STD-704, Category "B." Perform both of these tests to determine compliance with 3.13.15. In addition, apply at least three (3) aperiodic spike waveforms between Pins J1-C and J1-D. The spike waveform, produced by a generator with a source impedance of 50 ± 50 ohms, shall satisfy the following requirements:

Open-circuit voltage: +600 volts peak
Risetime: 0.9 ± 0.2 microseconds
Falltime: 10.0 ± 1.0 microseconds

4.10.16 Headset amp, unterminated. With input power applied and input signals removed, disconnect the 8 ohm load across pins J2-D and J2-E. Monitor the amplifier output each time the switches and volume controls are operated. Short pins J2-D and J2-E to ground for 1 minute. Observe that no unusual emanations occur from the unit while shorted and that no damage is sustained when the short is removed, to determine compliance with 3.13.16.

4.10.17 Transmitter control.

4.10.17.1 Transmitter control, 1 amp. Jumper pins J1-c and J1-d, set selector switch to position 1, and apply a plus dc voltage through a current limiting device to pin J1-f (negative to pin J1-D) to obtain a 1.0-ampere current flow. In turn, set selector switch to positions 2 through 5 as the dc voltage is applied to pins J1-g, J1-h, J1-r and J1-t. These current measurements shall be made to determine compliance with 3.13.17.

4.10.17.2 Transmitter control, GO/NO GO. Follow procedure of 4.10.17.1 except a GO/NO GO test may be made using a low current source. Connect a low current +28 volt source separately to all of pins JI-f, g, h, r, t and place selector switch in RMT. Remove jumper from J1-d to ground. Momentarily ground pin J1-B and note that there is automatic selection of the next radio channel as evidenced by voltage at next pin going to less than 1 volt when J1-d is grounded. Momentarily ground J1-B

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

enough times to scan through all 5 channels. Remove power from J1-t and then momentarily ground J1-B sufficient times to note that the next channel selected after channel 4 is channel 1. Continue to remove power from each pin in turn noting that the remote selection procedure skips all channels of which the control pins are not powered. When power is removed from all pins channel 1 will be selected. When power is removed from the channel 1 pin and power is applied to the channel 3 pin (or any other pin) the channel corresponding to the powered pin is selected.

Now jumper J2-h to ground and ground J2-Z momentarily. Note that channel 1 is selected only when J2-Z is grounded. Then ground J2-T and note selection and keying of channel 2. Similarly test channels 3, 4, 5 by momentarily grounding J1-a, F, B respectively.

4.10.18 Function selector logic.

4.10.18.1 Function selector logic, .5 amp. Jumper pins J1-c and J1-A, set selector switch to PVT, and apply a plus dc voltage through a current limiting resistor to pin J1-z (negative to pin J1-D) to obtain a 0.5-ampere current flow. Jumper pins J1-c and J1-d. In turn, set selector switch to positions ICS, and 1 through 5 as the dc voltage is applied to pins J1-Y, J1-Z, J1-j, J1-m and J1-n. These current measurements shall be made to determine compliance with 3.13.18.

4.10.18.2 Function selector logic, GO/NO GO. Follow procedure of 4.10.18.1 except a GO/NO GO test may be made using a low current source. Connect a low current +28 volt source separately to all of pins J1-f, g, h, r, t and place selector switch in RMT. Momentarily ground pin J1-B and note that there is an automatic selection of the next channel as evidenced by voltage at the next of pins J1-Z, j, j, m, n going to less than 1 volt. Momentarily ground J1-B sufficient times to scan through all 5 channels to verify proper operation.

Now jumper J2-h to ground and momentarily ground J2-Z, T, J1-a, F, B in turn to see that channels 1 through 5 are selected respectively as evidenced by voltage at J1-Z, j, j, m, n going to less than 1 volt respectively only when the corresponding control is momentarily grounded.

4.10.19 Microphone power. Set microphone selection switch in position 2, attach a 1000 ohm resistive load to pins J1-L and J1-M. Measure the dc voltage across the resistive load to determine compliance with 3.13.19.

4.10.20 Received audio input attenuation during transmit. Set selector switch to position 1, all monitor switches to ON and fully clockwise position, and HOT MIKE switch to NORM. Apply a 0.39 volt signal input at 1.0 kHz to pin pair J2-g and J2-P. Obtain 1.25 volts ± 10% at pin pair J2-D,E. Apply a ground to pin J1-d and measure the attenuation of the headset amplifier output at pins J2-D and J2-E with respect to the normal 1.25 volt output. In turn, apply the proper input, (0.39 volts to J2-R and J2-S, J2-g and J2-P) (2.75 volts to all other audio inputs), to all audio input pairs. Measure the headset amplifier output attenuation, apply a ground to pin J1-d and repeat this test for selector switch positions 1 thru 5 described in Table II to determine compliance with 3.13.20. Connect a jumper between pin J1-F and J1-c (key COMM). Perform tests again. This time 12 dB attenuation should occur even on selected channels.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

4.10.21 Audio level (volume adjustments).

4.10.21.1 Main volume control. Connect a 1.0 kHz input signal successively to each controlled received audio input (i.e., PVT, ICS, 1, 2, 3, 4, 5, NAV 1, NAV 2, DIR) and adjust the level to obtain 1.25 volt output across J2-D and J2-E, with both the individual monitor (where applicable) and master audio level controls set for maximum audio output. Adjust the main volume control for minimum audio output and measure this voltage for compliance with 3.13.21.1.

4.10.21.2 Monitor audio level adjustment. Connect a 1.0 kHz input signal successively to each switched received audio input (i.e., 1, 2, 3, 4, 5, NAV 1, NAV 2), and adjust the level to obtain 1.25 volt output across J2-D and J2-E, with both the individual monitor and master audio level controls set for maximum audio output. Adjust the appropriate individual audio level control for minimum audio output and measure this voltage for compliance with 3.13.21.2.

4.10.22 TEMPEST testing. The equipment shall be tested in accordance with the procedures of Appendix A and the contractor prepared, Government approved TEMPEST Test Plan.

4.11 Electromagnetic interference. The equipment shall be tested for compliance with the requirement of 3.14. Test setups and measurements techniques shall be in accordance with MIL-STD-462, Notice 2, as detailed in a contractor prepared, Government approved test plan. The unit(s) controls shall be adjusted as defined except as may be required to simulate special conditions. Selector switch to ICS, Hot MIC switch to Hot MIC, all monitor switches in ON, and MIC switch to position 1 (5 ohms), volume controls full clockwise position. The length of interconnecting cable shall be approximately six (6) feet. All emission tests shall be conducted with maximum input/output signal processed through the audio channels.

4.11.1 Audio susceptibility, CS 01. The unit(s) shall be tested in accordance with Method CS 01. The interphone output J2-R/J2-S shall be monitored with a scope and any deviation from normal output ±3 dB shall be recorded. In addition to the above, the noise output on headset and interphone lines shall be measured. These tests and measurements shall be made to determine compliance with 3.14.1.

4.11.2 Conducted susceptibility, CS 02. The unit(s) shall be tested in accordance with Method CS 02. The requirements of 3.14.2 shall be met.

4.11.3 Radiated susceptibility, RS 03. The unit(s) shall be tested in accordance with Method RS 03. The requirements of 3.14.3 shall be met.

4.11.4 Radiated susceptibility (induction field). Set the unit(s) to be tested six (6) inches directly above a wire which shall be placed parallel to the front of the unit and 1.25 inches behind the front panel. Connect the wire to a 400 Hz power source and adjust the current to 5 amps, rms. Set the unit controls as in 4.10.2 and 4.10.3 and measure the output levels to determine compliance with 3.14.4.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:19Z
Check the source to verify that this is the current version before use.

4.12 Interchangeability. The mechanical dimensions and electrical performance of parts shall be physically and electrically interchangeable among the unit(s). Mechanical dimensions and electrical performance required by the gauge drawings and equipment drawings shall be measured to determine conformance with physical and functional interchangeability requirements of 3.19. When a listed dimension or performance requirement is not within specified limits, it shall be considered a major defect. The following dimensions or performance shall be gauged or measured.

a. Mechanical (dimensions). The overall dimensions and mounting dimensions shall apply.

b. Electrical performance (assemblies). Shall be tested to determine compliance with the applicable performance requirements specified on the equipment drawings.

4.13 Operational inspections.

4.13.1 Operational inspections. Each unit(s) shall be connected to form an operational system. A talk test shall be performed to determine compliance with 3.18.

4.13.2 Operational inspection (system simulation). Each unit(s) shall be connected with at least nine other units (C-11746(V)( )/ARC) to form an operational system. (Substitution of simulated additional units is permissable when operational conditions are accurately duplicated. The substituted fixture shall be approved by the procuring activity.) A talk test shall be performed to determine compliance with 3.19.

4.14 Mechanical and visual inspections. The unit(s) shall be examined to determine conformance to the requirements of this specification including markings; accuracy of dimensions; quality of workmanship; maximum weight allowed; use of proper materials, parts, and processes; freedom from visible defects or any other imperfections that would result in rejection of the unit; and requirements 1, 5, 7, 8 and 9 of MIL-STD-454. All moving parts shall be examined to ensure that they operate freely without sticking or binding, and yet fit with the accuracy required by this specification and the applicable drawing. Defects shall be classified in accordance with MIL-STD-252.

4.14.1 Weight. The unit(s) shall be weighed to determine compliance with 3.2.1.

4.15 Quality conformance inspection of packaging. Packaging shall be tested in accordance with MIL-P-116 to determine conformance to the requirements of Section 5. The rough handling test will be performed only when specified (see 6.2c).

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or Commercial, as specified in contract (see 6.2b).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

5.1.1 Level A.

5.1.1.1 Cleaning. Control Unit, Communication System C-11746(V)( )/ARC shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Control Unit, Communication System C-11746(V)( )/ARC shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservative application. None required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be unit packed method IC-1 of MIL-P-116.

5.1.1.4.2 Packaging. Control Unit, Communication System C-11746(V)( )/ARC shall be individually unit packed Method III of MIL-P-116 as follows: Cushion the unit on all surfaces with cells or pads, or both, fabricated of fiberboard conforming to PPP-F-320, Type CF, class weather-resistant, variety SW, Grade W5c, designed to protect all projections and absorb the shock of impact in handling and transit. As an alternate, cushioning material may be fabricated of plastic molding material confonming to MIL-P-19644 or polyurethane form conforming to MIL-P-26514. Place each cushioned unit within a close-fitting fiberboard box conforming to PPP-B-636, W5C. Place the technical literature, unit packed as specified in 5.1.1.4.1, on top of the contents directly below the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level B. Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.1.

5.1.3 Commercial preservation. Preservation shall be in accordance with ASTM D 3951-82.

5.2 Packing. Packing shall be Level A, B or Commercial, as specified (see 6.2b). Shipping containers for Level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. A quantity of Control Unit, Communication System C-11746(V)( )/ARC, unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, Class 2, or PPP-B-585, style 2 or 3, Class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, Type I, Finish A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

5.2.1.1 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers, packed as specified in 5.2.1, except that container strapping may be omitted, shall be placed on a pallet, load Type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, Type IV, Group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, Type I, Finish A, or shrink film.

5.2.2 Level B. A quantity of Control Unit, Communication System C-11746(V)( )/ARC, unit packed as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film. When a single unit is shipped to a single destination no packing is required; the unit container shall serve as the shipping container. Reinforcing shall be as specified above.

5.2.2.1 Unitization. Palletization shall be required when quantities per destination comprise a pallet load. A quanity of Control Unit, Communication System C-11746(V)( )/ARC, unit packed as specified in 5.1, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, of shrink film.

5.2.3 Commercial packing. Packing shall be in accordance with ASTM D 3951-82.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with ASTM D 3951-82.

6. NOTES.

6.1 Intended use. Control Unit, Communication System C-11746(V)( )/ARC is intended for use by the pilot, copilot or crew to provide control of radio

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

equipments for voice transmission and reception, direction finding equipment and intercommunication when installed in aircraft and will provide remote transmit selection. When remote transmit selection is to be used the Indicator, Remote ID-2403/ARC may be installed to display the channel selection.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level A, level B or Commercial preservation and packing (see section 5).

c. When rough handling is required.

d. First article inspection:

(1) Several first article unit(s) are generally required so that lengthy environmental tests can be completed on specified sample units while complete performance measurements can be made on one sample unit (see 3.3).

(2) First article pack(s) as follows:

Makeup of pack(s).

Number of each kind of pack to be submitted.

Inspection to be performed thereon (including rough handling).

e. Marking and shipment of sample of first article samples and production unit(s).

f. Place of final inspection.

g. Submission of a statement of treatment per the requirements of paragraph 3.10.

6.3 Group C inspection. Approval to ship unit(s) on order may be withheld, at the discretion of the Government pending decision from the procuring activity, until compliance with 4.5.4.1 is complete.

6.4 Nomenclature. The nomenclatures assigned to the equipment covered by this specification are as follows:

Control Unit, Communication System C-11746(V)1/ARC (28V Red)

Control Unit, Communication System C-11746(V)2/ARC (5V Red)

Control Unit, Communication System C-11746(V)3/ARC (115V Green)

Control Unit, Communication System C-11746(V)4/ARC (28V Green)

Control Unit, Communication System C-11746(P)/ARC

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:19Z
Check the source to verify that this is the current version before use.

Panel, Illuminated, MX-10844/ARC (28V Red)

Panel, Illuminated, MX-10845/ARC (5V Red)

Panel, Illuminated, MX-10846/ARC (115V Green)

Panel, Illuminated, MX-10847/ARC (28V Green)

6.5 Deviations. Request for deviations in packaging and packing design shall be supported by test data proving the ability of the equipment, packaged and packed in accordance with the proposed design change, to withstand the applicable tests for the selected method, including the rough handling tests in accordance with MIL-P-116. Deviation approval shall be contingent on availability of packaging and packing materials selected and economics realized (see Section 5).

6.6 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.7 Definitions.

6.7.1 Branching. Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem or filament (hyphae).

6.7.2 Growth colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

6.7.3 Microbial growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

6.7.4 Sporulation. Sporulation is the formation of minute unicellular reproductive or dormant bodies, called spores.

6.7.5 Tubular germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

6.8 Subject term (key word) listing.

Intercommunication.

Custodian:

Army - AV

Preparing Activity:
Army - AV
(Project No. 5895-A324)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

APPENDIX A

TEMPEST REQUIREMENTS

FOR THE

CONTROL UNIT, COMMUNICATION SYSTEM C-11746(V)( )/ARC

NOTE: Appendix A, entitled as above, is classified CONFIDENTIAL and is enclosed under separate cover, as required.

APPLICABLE DOCUMENTS

The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

MIL-STD-831	Preparation of Test Reports
NACSEM 5100	Compromising Emanations Laboratory Test Standard, Electromagnetics (Document CONFIDENTIAL, Title Unclassified)

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both title and number or symbol should be stipulated when requesting copies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:19Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

Headquarters
U.S. Army Avionics R&D Activity
ATTN: SAVAA-C
Ft. Monmouth, NJ 07703-5401

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
U.S. Army Avionics Research &
Development Activity
ATTN: SAVAA-C
Fort Monmouth, NJ 07703-5401

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions -- Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-58111 (AV)	Control Unit, Communication System C-11746(V)( )/ARC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.